

Customer Segmentation & Reward Strategy

Project Summary

Data Science · June 2026

EXECUTIVE SUMMARY

TravelTide is an online flight-and-hotel platform with a large browsing audience but a comparatively small booking base — **65.5% of all sessions end without a booking**. The mandate for this project was to **segment the customer base by behaviour and design a tailored rewards programme** that converts browsers and turns occasional bookers into loyal, repeat customers.

Working from four raw platform tables (users, sessions, flights, hotels) spanning a 29-week window, we engineered a behavioural dataset of **46,863 sessions across 5,746 customers**, distilled 57 candidate features down to **19** across five behavioural dimensions, and applied **unsupervised two-stage K-Means clustering** — letting the data determine how many customer types exist rather than fixing it in advance. The result is **six actionable segments**, each paired with a reward whose mechanism is matched to that segment's dominant behaviour and its specific churn risk.

Segment	Share	Defining behaviour	Headline reward
Engaged mainstream bookers	37.8%	Active, multi-product core; books & bundles often, rarely cancels	Loyalty / points programme + bundle free-night
Light / occasional mainstream	37.5%	Same profile at lower intensity; lower spend, mostly solo	Time-limited “comeback” offer + small sweetener
Premium far-advance planners	7.7%	Highest fares & spend; book ~145 days ahead; international	No cancellation fees + premium service
Browse-only non-bookers	7.0%	Engage but never convert; intent present, barrier blocking	Bold first-booking offer + risk removal
International long-stay hotel	5.3%	Book hotels abroad but flights elsewhere; long stays	Bundle free-night to win the flight leg
High-engagement budget hunters	4.7%	Most engaged & price-driven; bundle most; cancel most (12%)	Members-only deals; reward completed trips

Key findings

- **Conversion, not loyalty, is the core gap** — two-thirds of sessions never book, so for some segments “retention” really means first-time activation.
- **Bundling is the single highest-leverage lever** — bundle-dominant customers spend 2.2× more, and the bundle motive recurs across three different segments.
- **The engaged mainstream carries the greatest absolute-revenue risk** — the largest segment (~\$4,800 lifetime spend each), so even small churn there is the most expensive churn.
- **Four niches are statistically distinct** (silhouette 0.50); the two mainstream tiers are a deliberate, business-driven split of one large continuum — stated transparently, not hidden.

Measurement & next step

To make the rewards programme provable, a retention KPI — the **90-day repeat-booking rate** — was designed and back-tested to baseline every segment, and a **six-arm, segment-stratified A/B test** is specified to establish, net of cost, which perks actually move retention before any company-wide roll-out. The recommended priorities: **protect the engaged mainstream first, make bundling a company-wide priority, and roll out each perk only once it has lifted repeat bookings, paid for itself, and not raised cancellations**.

DETAILED REPORT · PART I — PROBLEM, DATA & METHOD

1. Objective & analytical approach

TravelTide attracts heavy browsing traffic but converts a minority of it, and even among bookers, comparison-shopping and long gaps between trips create churn risk. The objective was a **behaviour-based segmentation** feeding a **tailored rewards programme** — the right incentive for the right customer to drive repeat, revenue-generating bookings. The method is deliberately **unsupervised**: the number and nature of segments are discovered from the data (k is not pre-set), and rewards are designed only after the segments are understood.

2. Data foundation

Four source tables were drawn from the TravelTide database and joined around the customer. The data is a single cross-sectional window, **4 Jan – 29 Jul 2023 (~29 weeks)** — a constraint the later measurement design explicitly works around.

Table	Grain	Key fields used
users	one row per customer	birthdate, gender, household, home location, sign-up date
sessions	one row per visit	start/end, page clicks, booking flags, discounts, cancellation, trip id
flights	one row per booked flight	seats, checked bags, base fare, route, airline, departure/return
hotels	one row per booked hotel	rooms, nights, price per room, check-in/out, hotel name

3. Data preparation

The recurring cleaning decision was separating **structurally missing** values (“the event never happened”) from genuine data-quality gaps:

- Blank trip, discount and return-time fields are **structural** (no booking, or a one-way flight) — kept or zero-filled, not treated as corrupt.
- Discount amounts are imputed to 0 where no discount was offered; where one was offered but the booking was cancelled, they are median-imputed from the matching completed population.
- 2,121 under-18 sessions** (4.3%) are removed — minors cannot legally book, and the cohort mixes genuine minors with birth-year typos.
- Hotel **nights** are recomputed from check-in/check-out dates (the raw column disagreed in 60.6% of rows); 227 rows with check-out before check-in are dropped. Final table: **46,863 sessions**.

4. Feature engineering & selection

Session-level signals were engineered first — booking type (browse / flight-only / hotel-only / bundle), session duration, booking lead time, total spend, discount depth, a completed-trip flag, and a careful three-layer **international-trip classifier** (handling Canadian codes and city-name collisions such as London-UK vs London-Ontario). These were aggregated to **5,746 customer profiles with 57 candidate features**, then reduced to **19** on four principles: independence (drop near-duplicates), rates over raw counts, completed-booking versions (to strip cancellation noise), and flight/hotel symmetry. The 19 span **booking behaviour, price & spend, travel profile, engagement, and customer lifecycle**, plus age (gender and home country were dropped as too skewed — 88% female, 83% USA — to separate customers).

5. What the data revealed (selected insights)

- Sessions rarely convert, and bundles are the value**: 65.5% browse-only; bundle-dominant customers (15% of users) spend a median \$5,233 vs \$2,424 — 2.2× more.
- Engagement is double-edged**: the most engaged quartile books 2.5× more (47% vs 19%) but also cancels far more (4.8% vs ~0%); engagement-to-cancellation correlation of 0.83 is the strongest link in the data.
- Two distinct trip types**: booking lead time is sharply bimodal — 95% book last-minute (~7 days, ~\$371) while a separate 5% book ~224 days ahead at ~\$1,775 (71% international), with almost nobody in between.
- No airline owns our customers**: only 24% of flyers stay loyal to one airline; 76% comparison-shop across two or more, and the more they spread, the more they spend.
- Never-bookers are a behavioural, not demographic, problem**: 402 customers browse but never buy, yet look demographically identical to bookers — a friction barrier, not the wrong audience.

DETAILED REPORT · PART II — SEGMENTATION & REWARD STRATEGY

6. Segmentation method — why two stages

Cluster quality (silhouette) stays healthy up to **k=5** and then falls off a cliff (0.429 at k=5 → 0.186 at k=6), so five is where genuine structure ends: four clean behavioural niches plus one **mainstream mass (75% of customers)**. That mass is too large and varied for a single perk, but the four niches are already distinct and left untouched. Re-clustering the mainstream alone shows no sharp internal gap — it is a soft continuum — so it is split into **two business-meaningful tiers (~50/50)**: an engaged tier and a light/occasional tier. Compressing skewed features with a log transform before clustering was decisive: it shrank the dominant mass from ~88% to 75% and let it divide into two near-equal, actionable halves.

7. The six segments

Segment	Size	Defining behavioural profile
Engaged mainstream bookers	2,173 (37.8%)	48% booking rate, 39% bundle, ~0% cancel, \$477 avg fare, \$4,794 lifetime spend, 3.1 airlines
Light / occasional mainstream	2,153 (37.5%)	27% booking, 19% bundle, \$360 fare, \$2,119 spend, mostly solo, fewer carriers (1.7)
Premium far-advance planners	440 (7.7%)	\$1,621 fare, 145-day lead time, 57% international, 6.3-night stays, \$7,868 spend (~4×
Browse-only non-bookers	402 (7.0%)	0% booking, \$0 spend, short sessions (11 clicks), highest recency — intent without conversion
International long-stay hotel	306 (5.3%)	100% international, 7.9-night stays at ~\$173/night, \$2,457 spend, no completed flights
High-engagement budget hunters	272 (4.7%)	43 clicks / 15-min sessions, 30% discount use, 46% bundle, 12% cancellation, \$409 fare

Validation. On their own, the four niches are compact and well-separated (silhouette 0.497, Davies-Bouldin 0.801). The full six-segment partition scores more modestly (silhouette 0.186) — by design, because the two mainstream tiers are a deliberate split of a continuum, not a natural gap the data offered.

8. Reward strategy

Rewards are drawn from a candidate set (free checked bag, free hotel meal, no cancellation fees, members-only discounts, a free hotel night with a flight, and loyalty points). The guiding rule is to **match the perk to the segment's behaviour and its specific churn risk** — never one-size-fits-all.

Segment	Recommended perk(s)	Retention mechanism
Engaged mainstream	Loyalty points + bundle free-night + free bag	Forfeitable, accumulating value locks in customers who are unattached, not unhappy
Light / occasional	Time-limited comeback offer + low-cost sweetener	Urgency shortens the booking gap; graduates them into the engaged tier
Premium planners	No cancellation fees + premium service	Flexibility neutralises long-lead-time risk; status loyalty protects the highest LTV
Browse-only	Bold first-booking offer + risk removal	Lowers the activation barrier; capped, test-and-learn budget
Intl long-stay hotel	Bundle free-night to win the flight leg	Captures share-of-wallet; converts a half-customer into a full-itinerary one
Budget hunters	Members-only deals; reward completed trips	Wins the constant price comparison while curbing the 12% cancellation rate

DETAILED REPORT · PART III — MEASUREMENT, LIMITATIONS & CONCLUSIONS

9. Retention KPI — the 90-day repeat-booking rate

Success is measured with one metric: **RBR-90**, the share of a segment that completes at least one non-cancelled booking within 90 days of a reference date. A landmark-plus-forward-window design lets the **same formula** profile a segment today and read an experiment later, so today's baseline is directly the control expectation. It is segment-adaptive: for browse-only customers, who have no prior bookings, RBR-90 simply becomes the first-booking (activation) rate. The historical back-test gives each segment a baseline and the supporting metrics needed to plan a fair test:

Segment	RBR-90	Active 90d	Bookings/user	Flight-attach	Revenue retained
Engaged mainstream	74.8%	94.7%	1.07	68.9%	\$1,380
Light / occasional	49.0%	91.3%	0.57	42.3%	\$582
Premium planners	73.1%	91.1%	0.88	73.1%	\$2,749
Browse-only (activation)	0.0%	85.5%	0.00	0.0%	\$0
Intl long-stay hotel	34.0%	88.6%	0.37	0.0%	\$802
Budget hunters	67.6%	91.9%	0.97	60.7%	\$1,009

10. Validation experiment (A/B test)

Because the perks pull in opposite directions, the plan is **six experiments, not one** — each randomised and read within its own segment, perk arm versus a no-perk holdout. The primary endpoint is RBR-90, with **cancellation rate and net revenue after perk cost** as guardrails. The decision rule is strict: ship a perk only if it **lifts RBR-90 significantly, covers its cost, and does not raise cancellations**. Because the customer base is fixed, the binding constraint is statistical power — the smallest detectable lift (MDE) per segment:

Segment	N	Baseline RBR-90	Detectable lift	With CUPED
Engaged mainstream	2,173	74.8%	±4.5 pp	±3.9 pp
Light / occasional	2,153	49.0%	±5.4 pp	±4.6 pp
Premium planners	440	73.1%	±9.8 pp	±8.6 pp
Browse-only	402	0.0%	±3.0 pp	±2.3 pp
Intl long-stay hotel	306	34.0%	±13.9 pp	±12.0 pp
Budget hunters	272	67.6%	±13.1 pp	±11.5 pp

The two mainstream segments are well-powered for small, realistic lifts; the small niches can only detect large effects, so they rely on variance reduction (CUPED) and are read more directionally on guardrail economics.

11. Limitations & trade-offs

- **Snapshot data** — a single 29-week window; retention is reconstructed via a historical back-test, and the 90-day window understates the 145-day-lead premium planners (read at 180 days instead).
- **The mainstream split is business-driven**, not a natural data gap — a transparent choice that makes that boundary softer than the niche boundaries.
- **Small niches are underpowered** for modest effects, even with variance reduction.
- **Perk economics are not yet quantified** — the net-revenue-after-cost guardrail is defined but per-perk margins must be supplied before the test reads as a profit decision.

12. Conclusions

- **Retention here is two problems.** For ~75% of customers it means deepening frequency and lock-in (loyalty + bundles); each niche has its own lever — flexibility, activation, share-of-wallet, or price credibility.
- **Bundling is the connective lever** — it drives 2.2× spend and routes through three segments' perks; making it a company-wide priority compounds value.
- **Protect the engaged mainstream first** — the largest, highest-revenue group, where churn is most costly in absolute terms.
- **Measurement is built in, not bolted on** — one KPI both profiles segments and reads the experiment, turning “this should work” into “this did work, net of cost.”